

# Delayed Market Making

June 18, 2026

At step  $t$ :

1. Market reveals  $P_t \in [0, 1]$ .
2. Learner picks  $a_t \in [0, 1]$ .
3. For all  $t' < t$ , if  $a_{t'} < P_t$ , then  $a_{t'}$  gets traded and the learner gets reward  $a_{t'}$ .

Let

$$\tau_t(a) := \min\{s > 0 : P_{t+s} > a\}.$$

Then the reward from posting quote  $a$  at time  $t$  is

$$r_t(a) = a \cdot \mathbf{1}\{\tau_t(a) \leq T - t\}.$$

The regret is

$$R(T) = \max_{a \in [0, 1]} \mathbb{E} \left[ \sum_{t=1}^T r_t(a) \right] - \mathbb{E} \left[ \sum_{t=1}^T r_t(a_t) \right].$$

**Assumption 1.** *Let*

$$\beta(h) := \sup_t \sup_a \text{ess sup}_{P_1, \dots, P_t} \Pr[h < \tau_t(a) < \infty \mid P_1, \dots, P_t].$$

*Assume*

$$\lim_{h \rightarrow \infty} \beta(h) = 0.$$

**Conjecture 1.** *Sublinear regret is possible if and only if Assumption 1 holds.*

*Proof. If direction.*

- Assume  $K$  arms. Later, we will control the error due to discretization. (Lemma 2)
- Pick some  $H(T)$  that is sublinear. We will optimize the exact form of  $H(T)$  for optimal regret.
- Run an algorithm for expert advice with  $K$  arms and delay  $H(T)$ . If a quote has not traded after  $H(T)$  steps, assume it will never trade.

- Consider the learning problem you get if the delay actually has an upper bound of  $H(T)$ , i.e., define

$$r_t^H(a) = a \cdot \mathbf{1}\{\tau_t(a) \leq \min\{H(T), T - t\}\}$$

and the corresponding regret

$$R^H(T) = \max_{a \in [0,1]} \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a) \right] - \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a_t) \right].$$

- From standard expert advice with delay algorithms, we get  $R^H(T) = O(\sqrt{TH(T) \log K})$
- (Lemma 1) For each fixed  $a$ , we have that

$$\mathbb{E} [r_t(a) - r_t^H(a)] \leq \Pr[H(T) < \tau_t(a) < \infty] \leq \beta(H(T)).$$

Since  $r_t(a_t) \geq r_t^H(a_t)$ , it follows that  $R(T) \leq R^H(T) + T\beta(H(T))$ , and therefore

$$R(T) \leq O(\sqrt{TH(T) \log K}) + T\beta(H(T))$$

- Thus under Assumption 1, we get sublinear regret. Furthermore, if we know the exact form of  $\beta(h)$ , then we can choose  $H(T)$  appropriately to balance the two terms.

Below are some details.

**Lemma 1** (Truncation comparison). *Let  $H = H(T)$ , and let*

$$r_t^H(a) = a \cdot \mathbf{1}\{\tau_t(a) \leq \min\{H, T - t\}\}.$$

*Fix any learner and let  $R(T)$  and  $R^H(T)$  be its expected regrets under the rewards  $r_t$  and  $r_t^H$ , respectively. Then*

$$R(T) \leq R^H(T) + T\beta(H).$$

*Proof.* Fix a quote  $a \in [0, 1]$ . The true reward  $r_t(a)$  and the truncated reward  $r_t^H(a)$  differ only when the quote trades before the end of the game, but more than  $H$  steps after it was posted. Thus

$$0 \leq r_t(a) - r_t^H(a) \leq \mathbf{1}\{H < \tau_t(a) < \infty\}.$$

Taking conditional expectations given the realized history  $P_1, \dots, P_t$ , we get

$$\mathbb{E} [r_t(a) - r_t^H(a) \mid P_1, \dots, P_t] \leq \Pr[H < \tau_t(a) < \infty \mid P_1, \dots, P_t] \leq \beta(H),$$

where the last inequality is exactly the definition of  $\beta(H)$ . Taking expectations and summing over  $t$ , this gives

$$\mathbb{E} \left[ \sum_{t=1}^T r_t(a) \right] \leq \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a) \right] + T\beta(H).$$

Since this holds for every fixed quote  $a$ ,

$$\max_{a \in [0,1]} \mathbb{E} \left[ \sum_{t=1}^T r_t(a) \right] \leq \max_{a \in [0,1]} \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a) \right] + T\beta(H).$$

On the other hand, along every realized path we have  $r_t^H(a_t) \leq r_t(a_t)$  for every  $t$ . Therefore

$$\mathbb{E} \left[ \sum_{t=1}^T r_t(a_t) \right] \geq \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a_t) \right].$$

Combining, we get,

$$\begin{aligned} R(T) &= \max_{a \in [0,1]} \mathbb{E} \left[ \sum_{t=1}^T r_t(a) \right] - \mathbb{E} \left[ \sum_{t=1}^T r_t(a_t) \right] \\ &\leq \max_{a \in [0,1]} \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a) \right] - \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a_t) \right] + T\beta(H) \\ &= R^H(T) + T\beta(H). \end{aligned}$$

□

**Lemma 2** (Discretization comparison). *Let  $K \geq 2$ , and let*

$$\mathcal{G}_K = \left\{ 0, \frac{1}{K}, \dots, \frac{K-1}{K} \right\}.$$

*Consider a grid-restricted problem where  $a_t \in \mathcal{G}_K$  for every  $t$ . Define regret of a learner as*

$$R_K^H(T) = \max_{b \in \mathcal{G}_K} \mathbb{E} \left[ \sum_{t=1}^T r_t^H(b) \right] - \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a_t) \right].$$

*Let  $R^H(T)$  be the regret of the same learner in the original problem, where the comparator may choose any fixed action in  $[0, 1]$ :*

$$R^H(T) = \max_{a \in [0,1]} \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a) \right] - \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a_t) \right].$$

*Then,*

$$R^H(T) \leq R_K^H(T) + \frac{T}{K}.$$

*Proof.* Let  $L_t = \min\{H, T - t\}$ . For every quote  $a \in [0, 1]$ , choose a grid point  $b(a) \in \mathcal{G}_K$  such that

$$b(a) \leq a \quad \text{and} \quad a - b(a) \leq \frac{1}{K}.$$

This is possible by rounding  $a$  down to the nearest point of  $\mathcal{G}_K$ .

Fix  $a$  and write  $b = b(a)$ . Since  $b \leq a$ , any future price that exceeds  $a$  also exceeds  $b$ . Therefore, if quote  $a$  trades within  $L_t$  steps, then quote  $b$  also trades within  $L_t$  steps. Equivalently,

$$\mathbf{1}\{\tau_t(a) \leq L_t\} \leq \mathbf{1}\{\tau_t(b) \leq L_t\}$$

on every realized path. Hence, pathwise,

$$\begin{aligned} r_t^H(a) &= a \cdot \mathbf{1}\{\tau_t(a) \leq L_t\} \\ &= b \cdot \mathbf{1}\{\tau_t(a) \leq L_t\} + (a - b) \cdot \mathbf{1}\{\tau_t(a) \leq L_t\} \\ &\leq b \cdot \mathbf{1}\{\tau_t(b) \leq L_t\} + (a - b) \\ &\leq r_t^H(b) + \frac{1}{K}. \end{aligned}$$

Taking expectations and summing over  $t$ , we get

$$\mathbb{E} \left[ \sum_{t=1}^T r_t^H(a) \right] \leq \mathbb{E} \left[ \sum_{t=1}^T r_t^H(b(a)) \right] + \frac{T}{K} \leq \max_{b \in \mathcal{G}_K} \mathbb{E} \left[ \sum_{t=1}^T r_t^H(b) \right] + \frac{T}{K}.$$

Since this holds for every  $a \in [0, 1]$ , the best continuum comparator is within  $T/K$  of the best grid comparator:

$$\max_{a \in [0, 1]} \mathbb{E} \left[ \sum_{t=1}^T r_t^H(a) \right] \leq \max_{b \in \mathcal{G}_K} \mathbb{E} \left[ \sum_{t=1}^T r_t^H(b) \right] + \frac{T}{K}.$$

Subtracting the learner's expected truncated reward  $\mathbb{E} \left[ \sum_{t=1}^T r_t^H(a_t) \right]$  from both sides yields

$$R^H(T) \leq R_K^H(T) + \frac{T}{K}.$$

□

**Putting the terms together.** The proof sketch above began by pretending that the problem already has  $K$  arms. Lemma 2 explains how to justify this: run the learner on the grid  $\mathcal{G}_K$ , and then compare its performance to the original comparator. If the delayed finite-arm algorithm guarantees

$$R_K^H(T) = O \left( \sqrt{TH \log K} \right),$$

then Lemmas 1 and 2 give

$$\begin{aligned} R(T) &\leq R^H(T) + T\beta(H) \\ &\leq R_K^H(T) + \frac{T}{K} + T\beta(H) \\ &= O \left( \sqrt{TH \log K} \right) + \frac{T}{K} + T\beta(H). \end{aligned}$$

Thus the regret bound has three terms:

$$\underbrace{O \left( \sqrt{TH \log K} \right)}_{\text{learning with delay}} + \underbrace{\frac{T}{K}}_{\text{discretization}} + \underbrace{T\beta(H)}_{\text{truncation}}.$$

The discretization term is explicitly  $T/K$ .

The parameters  $K$  and  $H$  should be chosen to balance these three terms. For a fixed  $H$ , the grid should be fine enough that  $T/K$  does not dominate the delayed-learning term. Since  $K$  appears only logarithmically in the delayed-learning term, a simple safe choice is  $K = T$ , which makes  $T/K = 1$  and gives

$$R(T) \leq O \left( \sqrt{TH \log T} \right) + T\beta(H) + 1.$$

A more economical choice, up to logarithmic factors, is to take  $K \approx \sqrt{T/H}$ , which makes the discretization term no larger than the delayed-learning term.

After  $K$  is chosen,  $H$  controls the tradeoff between the delayed-learning term and the truncation term. Increasing  $H$  makes the finite-arm delayed learning problem harder, because the term

$\sqrt{TH \log K}$  grows with  $H$ . But increasing  $H$  also makes the truncation error  $T\beta(H)$  smaller. The best choice is therefore obtained by approximately solving

$$\sqrt{TH(T) \log K(T)} \approx T\beta(H(T)).$$

Equivalently, with the simple choice  $K = T$ , one balances

$$\beta(H(T)) \approx \sqrt{\frac{H(T) \log T}{T}}.$$

There is no universal optimal  $H(T)$  without more information about the decay of  $\beta$ . If  $\beta(h) \lesssim h^{-\alpha}$ , then the balancing choice with  $K = T$  is

$$H(T) \asymp \left( \frac{T}{\log T} \right)^{1/(2\alpha+1)},$$

which gives regret of order

$$T^{(\alpha+1)/(2\alpha+1)} (\log T)^{\alpha/(2\alpha+1)}$$

up to constants. If instead  $\beta(h)$  decays exponentially, then  $H(T)$  only needs to grow logarithmically in  $T$ , giving regret nearly  $\sqrt{T}$  up to logarithmic factors.

For the qualitative sublinear-regret result, Assumption 1 is enough. Choose  $H(T) \rightarrow \infty$  slowly enough that  $H(T) \log T = o(T)$ , and take  $K = T$ . Then  $\sqrt{TH(T) \log T} = o(T)$ ,  $T\beta(H(T)) = o(T)$ , and  $T/K = 1$ .

*Only if direction.* □